MIL-C-0083933A(MR)
5 October 1970
USED IN LIEU OF
MIL-C-83933, Grade 1
29 September 1967
(See 6.5)

MILITARY SPECIFICATION

CORROSION PREVENTIVE COMPOUND, COLD-APPLICATION (FOR MOTOR VEHICLES)

This limited coordinated Military Specification has been prepared by the Army Materials and Mechanics Research Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-C-83933. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers a solvent dispersed corrosion preventive compound for spray, brush, or dip application to vehicle underbodies and inclosed or concealed surfaces (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-M-232 - Methanol (Methyl Alcohol)
QQ-S-698 - Steel, Carbon, Sheet and Strip
TT-N-95 - Naphtha, Aliphatic

STANDARDS

FEDERAL

FED-STD-791 - Lubricants, Liquid Fuels and Related Products, Methods of Testing

FCS 8030

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290 - Packaging, Packing, and Marking of Petroleum Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

D 92 Flash and Fire Points by Cleveland Open Cup

D 95 Water in Petroleum and Other Bituminous Materials

D 130 Copper Corrosion by Petroleum Products

D 874 Sulfated Ash from Lubricating Oils and Additives

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical Society specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The corrosion preventive compounds furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5.1 and 6.3).

3.2 Material. The corrosion preventive compound shall be composed of a grit and abrasive free nonvolatile base material dispersed in a petroleum solvent so as to provide a homogeneous formulation which will meet the specified requirements.

3.3 Physical and chemical requirements. The compound shall conform to the physical and chemical requirements specified in Table I and 3.4 through 3.15.

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Table I

Requirement	Value
Flash point, °F, min:	100
Fire point, °F, min:	105
Water content, percent, max:	1.0
Copper strip corrosion (3 hours at 212°F) maximum ASTM classification:	2e

3.4 Condition in container. The compound shall show no settling in a freshly opened, full container, no evidence of a separation of solvent, and shall be free from lumps or skins (see 4.6.1).

3.5 Nonvolatile content. The nonvolatile content, percent by weight, for the compound will be established when qualification tests are conducted, but shall not be less than 52 percent based on the compound as received as being 100 percent. The nonvolatile content of any succeeding lot of the compound shall be no lower, and not more than 5 percent higher than the established value (see 4.6 and 6.4).

3.6 Sulfated residue. The sulfated residue of the compound will be established when qualification tests are conducted. When the established value of the accepted sample is 0 to 0.50 percent, the sulfated residue of any succeeding lot shall be ± 0.05 (absolute value) with a zero minimum. When the established value of the accepted sample is 0.51 percent or higher, the sulfated residue of any succeeding lot shall be ± 15 percent of the established value (see 4.6.1). The deviations permitted herein are manufacturing tolerances and are separate from those indicated in 4.6. The latter deviations reflect the accuracy of the method (see 4.6 and 6.4). Lead shall not be present.

3.7 Halogen content. The compound shall not contain any halogenated hydrocarbons (see 4.6.2) or other solvents of a highly toxic nature (see 6.3.1).

3.8 Fire resistance. The compound, when applied to panels and air-dried for seven days, may char but shall not support combustion for more than 15 seconds after the flame source is removed when tested as specified in 4.6.3.

3.9 Low temperature stability. The compound shall show no evidence of separation or settling out at the completion of the storage period (see 4.6.4).

3.10 Low Temperature flexibility. The compound, when applied to panels and air-dried for seven days, shall not crack, peel or chip when tested in accordance with 4.6.5.

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3.11 High temperature flow. The compound, when applied to panels and air-dried for seven days, shall not flow at a temperature below 225°F (107°C) (see 4.6.6). Failure shall be considered as any movement of the coating toward the reference line.

3.12 Corrosion protection.

3.12.1 Salt spray resistance. After a minimum of 500 hours exposure in a salt spray cabinet operated using a 5-percent sodium chloride solution, not more than three corrosion dots, none of which exceed one millimeter in diameter, shall be evident on any of the test panels. For the panel with the scribe mark, the compound shall show no signs of undercutting greater than 1/32 inch on either side of the scribe mark. Corrosion on the outer 1/4 inch of the panel shall not be cause for rejection (see 4.6.7.1).

3.12.2 Salt water immersion. After a minimum of 14 days immersion in a synthetic sea water solution, the coated test panels shall show no evidence of attack by the sea water. After removal of the coating with solvent, not more than three corrosion dots, none of which exceed one millimeter in diameter, shall be evident on any of the test panels. Corrosion on the outer 1/4 inch of the panel shall not be cause for rejection (see 4.6.7.2).

3.13 Creep. After a 7-day test period, the compound shall show evidence of creep of 1/4 inch or more between the test panels when tested in accordance with 4.6.8.

3.14 Sprayability. When tested as specified in 4.6.9, the compound shall be considered sprayable if the film on the sprayed glass panel is continuous and uniform with no sagging or build-up on the lower part of the panel.

3.15 Condition to touch. The compound, when applied to a panel and air-dried for seven days, shall be considered dry to touch when firm pressure with the tip of the finger shows a slight tacky condition without any coating adhering to the finger. There shall be no evidence of rupture of the film (see 4.6.10).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container; or manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

4.2.2 Packaged lot. An indefinite number of 55 gallon drums or smaller unit packages of identical size and type, offered for acceptance, and filled with a homogeneous mixture of material from one isolated container; or filled with a homogeneous mixture of material manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

4.3 Sampling.

4.3.1 Sampling for examination of packed containers. Take a random sample of packed containers from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTM Method D 270.

4.4 Inspection. Perform inspection in accordance with Method 9601 of Fed. Test Method Std. No. 791.

4.4.1 Examination of packed containers. Examine samples taken in accordance with 4.3.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests.

4.5.1 Qualification tests. Qualification tests shall consist of all the tests of this specification.

4.5.2 Quality conformance tests. Quality conformance tests shall consist of all the tests of this specification except salt spray resistance (see 4.6.7.1) and salt water immersion (see 4.6.7.2).

4.6 Test method. Perform tests in accordance with the applicable test methods given in Table III and in paragraphs 4.6.1 through 4.6.10 as applicable.

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Table III. Test Methods

Test	Test method no. Fed. Std. 791	Test method no. ASTM
Flash and fire point		D 92
Water content		D 95
Copper strip corrosion		D 130
Nonvolatile content	3480	
Sulfated residue		D 874

4.6.1 Condition in container. The condition of the material shall be observed in the container in which the material was submitted for test and prior to any agitation. The presence of lumps or skins shall be noted and a spatula or paddle shall be lowered into the container to establish evidence of separation or settling.

4.6.2 Halogen content. A small loop of 18 to 20 gage copper wire shall be heated in the flame of a Bunsen burner until there is no color to the flame. The loop shall be cooled and dipped into a sample of the compound under test. The loop is placed in the outer portion of the flame and after the luminous flame has disappeared, observe the flame for the presence of a green coloration due to the presence of halogenated compound.

4.6.3 Fire resistance. Three test panels of the size and composition and cleaned as specified in 4.6.7.1 shall be used for this test. A coating of the compound shall be applied to one side of the panels by means of a doctor blade which has been adjusted to provide a clearance of 0.012 ± 0.001 inches. The panels shall be permitted to air-dry for 7 days ± 2 hours at 77° ± 5°F (25° ± 2°C). One of the test panels shall be suspended vertically in a shielded hood. The panel shall be placed in the flame of a Bunsen burner with the air shut off and the flame regulated to 2 ± 1/4 inches under the panel so that the lower end of the panel is in the flame 1 ± 1/8 inch. The flame shall be allowed to remain under the test panel for 20 ± 1 seconds. The flame shall then be withdrawn and the time that flaming continues shall be observed. The test shall be run in duplicate and the results averaged.

4.6.4 Low temperature stability. A pour point jar shall be filled to the mark with the compound and stoppered. The jar shall then be placed in a suitable cold chamber maintained at -20° ± 1°F (-29° ± 0.5°C) and allowed to remain at this temperature for 16 hours ± 15 minutes. Care should be taken to avoid physical disturbance of the compound. The sample shall then be removed from

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the cold chamber and allowed to remain at 77° ± 5°F (25° ± 2°C) for 8 hours ± 15 minutes. This cycle shall be performed for a total of four times, ending with the expiration of the last exposure at -20° ± 1°F (-29° ± 0.5°C). The jar shall be placed at 77° ± 5°F (25° ± 2°C) for six hours ± 10 minutes after which the tube shall be inverted six times and permitted to stand at 77° ± 5°F (25° ± 2°C) for one hour. Examination shall then be made for compliance with 3.9.

4.6.5 Low temperature flexibility. Three panels, 2 by 5 inches, shall be cut from cold-rolled, No. 21 gage, rust-free steel conforming to Specification QQ-S-698. The steel shall have a Rockwell "B" hardness of between 55 and 65. The panels shall be cleaned as specified in 4.6.7.1.1 and coated and conditioned as specified in 4.6.3. The panels shall be permitted to air-dry for 7 days ± 2 hours at 77° ± 5°F (25° ± 2°C). The panels shall then be conditioned at -10° ± 2°F (14° ± 1°C) for 2 hours ± 5 minutes. While at that temperature, each test panel shall be bent quickly 180 degrees around a mandrel 3/16 inch in diameter which has been cooled to -10° ± 2°F (14° ± 1°C). Plastic or rubber gloves shall be used to handle the panels and no more than 5 seconds shall elapse from the time the chamber is opened and the panel is bent. The panel shall then be hammered several times with the ball end of a 1-pound ball peen hammer and observed for compliance with 3.10.

4.6.6 High temperature flow. Three test panels, 2 by 4 by 1/8 inch shall be of the same composition and cleaned as specified in 4.6.7.1 and coated and conditioned as specified in 4.6.3. Using a stiff spatula, knife, or razor blade, remove a one-inch strip of the coating along one of the long edges of the panel, after cutting the film parallel to the edge of the panel, and at a right angle to the surface of the panel. A straight line shall then be drawn or scratched on the panel, parallel to the cut edge of the film, and 1/8 inch away from it. The panel shall then be suspended vertically, with the exposed area downward, for 2 hours in a gravity convection oven maintained at 225° ± 2°F (107° ± 1°C). The panels shall be removed from the oven and allowed to cool to 77° ± 5°F (25° ± 1°C). The position of the cut edge of the coating relative to the reference line drawn, or scratched on the panels shall then be observed for compliance with 3.11.

4.6.7 Corrosion protection.

4.6.7.1 Salt spray resistance.

4.6.7.1.1 Test panels. Three panels for the salt spray resistance test shall be the same size and composition and initially cleaned as specified in Method 5329 of FED-STD-791. The panels shall be polished to a surface finish of 15 to 20 microinches (RMS) with 150 grit silicon carbide or aluminum oxide abrasive followed by 240 grit abrasive.

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4.6.7.1.2 Test procedure. The panels shall be coated and conditioned as specified in 4.6.3. Using a knife, a scribe mark shall be made lengthwise in the middle of the test surface of one of the panels. The back side and edges of all three panels shall be sealed with an appropriate inert material or a thick coating of the material under test. The panels shall then be exposed for 500 hours using a 5 percent sodium chloride solution in accordance with Method 4001 of FED-STD-791. At the completion of the exposure period, the panels shall be rinsed with water and then methyl alcohol conforming to O-M-232. The compound shall be removed with naphtha conforming to TT-N-95 and panels then given a final rinse in methyl alcohol. The panels shall then be examined for compliance with 3.12.1.

4.6.7.2 Salt water immersion. Three test panels, 2 by 3 by 1/16 inch, of the same composition and cleaned as specified in 4.6.7.1.1 shall be coated and conditioned as specified in 4.6.3. Seal the edges and back side of the panels with any appropriate inert material.

4.6.7.2.1 Preparation of the salt water. The salt water shall be prepared with chemicals conforming to ACS standards for analytical reagent chemicals as follows:

Ingredients	Grams per liter of solution
Sodium Chloride	27.6
Calcium Chloride	2.4

The pH of the solution shall be adjusted to a value of 7.8 to 8.2 by addition of 5 percent solution of sodium carbonate. If the salt water solution is kept in stock it shall be checked, and its pH adjusted to the specified value, prior to each test.

4.6.7.2.2 Test procedure. The coated and conditioned test panels shall be separately immersed in a vertical position for 14 days ± 2 hours in 500 ml tall-form beakers containing salt water to a depth of 3/8 to 5/8 inch above the panels. The panels shall be supported in a vertical position by inserting them in slots in cylindrical hardwood blocks, 2-1/8 inches in diameter and 1 inch in height. Each slot shall be 1/8 inch deep and 1/16 inch wide. The blocks shall have been pre-conditioned prior to initial use by immersion for 48 hours in the salt water. The beaker shall be covered with a watch glass. The temperature of the salt water during immersion shall be 77° ± 5°F (25° ± 2°C). Additions of salt water shall be made throughout the test to maintain the proper level of salt water. At the completion of the test the panels shall be removed from the salt water and the coatings examined for evidence of attack. The panels shall then be rinsed with water and methyl alcohol. The compound shall be removed with naphtha and panels rinsed in methyl alcohol. Examine panels for compliance with 3.12.2.

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4.6.8 Creep. Two test panels, 2 by 5 inches, of the same gage and composition as specified in 4.6.5 shall be cleaned as specified in Method 5329 of FED-STD-791. The ends of the two panels shall be fastened together with a 1/2 inch overlap by means of two 1/8 inch steel blind rivets (maximum shear strength 325 pounds) centered uniformly in the 1/2 inch overlap area. The rivets shall be mechanically expanded by a pull stem. The compound shall be applied to the top edge of the joint by means of a spatula. The panels shall remain in a vertical position for 7 days ± 2 hours at 77° ± 5°F (25° ± 2°C). Separate the panels with care to avoid disturbing the compound between them. Examine the separated panels for compliance with 3.13.

4.6.9 Sprayability. A sample of the compound shall be tested for sprayability at 40°F and 100°F. A sample of the compound in a tightly sealed container shall be conditioned at 40 to 45°F for 24 hours. An airless type spray gun (Alemite No. 8596 or equal) fitted with a spray nozzle having an orifice of 0.015 ± 0.001 inch designed to produce a fan shaped pattern shall be used for the sprayability test. Condition the spray gun at 40 to 45°F for a minimum of two hours. The material shall be sprayed on a glass panel of a minimum size of 10 by 10 inches. The panel shall be placed in a vertical position with the spray gun held horizontally and approximately 12 inches from the panel. The compound shall be sprayed to a minimum dry film thickness of 3 mils. The panel shall be held in a vertical position at room temperature and examined after 1 hour for compliance with 3.14. The sprayability test shall be repeated at 95 to 100°F.

4.6.10 Condition to touch. A 2 by 4 by 1/8 inch panel of the same composition and cleaned as specified in 4.6.7.1.1 shall be coated and conditioned as specified in 4.6.3. The panel shall be permitted to air-dry for 7 days ± 2 hours at 77° ± 5°F (25° ± 2°C). The coating shall then be tested at points not less than 1/2 inch from the edge. The test shall be made by applying the fingertip firmly to the compound and observing for compliance with 3.15.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with MIL-STD-290. The type and size of containers and level of packaging and packing shall be as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The corrosion preventive compound described in this specification is intended for use in preserving the underside and internal areas

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of vehicles. Preserved areas that are particularly susceptible to abrasion may require periodic touch-up. The compound can be effectively used over previously applied underbody coatings if these old coatings are tightly adhering to the metal.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Quantity.

(c) Type and size of containers.

(d) Level of packaging and packing.

(e) Any special markings.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is Hq., U. S. Army Weapons Command, ATTN: AMSWE-RET-ML, Rock Island, Illinois 61201, and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Any questions raised regarding toxicity should be referred by the procuring agency to the departmental medical authority. For Army procurement, the Surgeon General will act as advisor to the Army.

6.4 Qualification approval test data on approved products required for conducting the sulfated residue test is available upon request from the U. S. Army Weapons Command, ATTN: AMSWE-RET-ML, Rock Island, Illinois 61201.

6.5 This interim specification may be used in lieu of Grade 1 of Specification MIL-C-83933. The grades 2, 3, and 4 of MIL-C-83933 are still available under MIL-C-83933 since that portion of the specification has not been cancelled.

6.6 Interim Military Specification MIL-C-0083933(MR) differs primarily from Specification MIL-C-83933 in the following respects:

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(a) Only one grade of corrosion preventive compound is covered instead of four grades.

(b) The coatings are applied to the test panels by a doctor blade rather than by spray gun.

(c) The sprayability requirement is specified at 40° and 100°F instead of 0° and 100°F.

(d) The water displacement requirement and test method have been eliminated.

(e) The activity responsible for the Qualified Products List was changed to show the Hq., U. S. Army Weapons Command instead of Robins Air Force Base.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
WC, MD, AT

Project No. 8030-A026

User activities:
MU, MI

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-C-0083933A(MR), Corrosion Preventive Compound, Cold Application (for Motor Vehicles)

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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Army Materials and Mechanics Research Center
Watertown, Massachusetts 02172

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